

Point-in-Time Audit Before Alpha: Public-Archive Availability and a Negative Matched-Budget Study on BTC Perpetual Futures

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Abstract

Public cryptocurrency archives may appear usable when files exist, although factor research requires observations available and executable at each decision time. We audit public Binance BTCUSDT USD-M perpetual-futures data using event, publication, and availability times and separate proposal from deterministic auditing, evaluation, and hold-out access. An initial gapless five-minute requirement for trade, mark, index, and open interest failed: the longest unrepaired intersection was **304.5729166666667 days**. A disclosed revision made trade, mark, index, and realized funding the core streams and made OI optional because its publication time was unverified. The revised mask retained **727 complete UTC days** and supported a mechanical 436/145/146-day train/validation/historical-holdout split.

On 80 frozen known-rule templates, the auditor detected 40/40 violations and rejected 0/40 legal templates; Wilson 95% intervals were 91.24%–100% and 0%–8.76%, so this is template-level evidence. Across ten null-signal paths, full auditing reduced mean false passes from 0.2910 to 0.0625 (78.5%; paired path-bootstrap difference -0.2285, 95% interval [-0.2595, -0.2012]). The simulation did not distinguish exact-grid deletion from availability masking and showed no material PBO improvement.

Under a shared DSL and matched valid-candidate budgets, a deterministic adaptive audited agent and random search each produced 39 validation-qualified candidates; tree GP produced 29. In the one-time historical holdout, all 23 evaluated runs had positive IC but negative net Sharpe under primary costs, and none of 460 fee-slippage cells had positive Sharpe. We retain the audit and null-control result but reject agent superiority, profitability, stability, and an independent masking effect.

Keywords: perpetual futures; point-in-time data; availability time; factor mining; false discovery; negative result

1. Introduction

Formulaic factor discovery is commonly presented as a search problem. For perpetual futures, however, a candidate can appear predictive because the underlying record was unavailable, the trade used an unfinished bar, realized funding was moved backward, or a gap was filled with future information. A valid study must therefore establish not only what an observation represents, but when it could have been used.

We study public BTCUSD USD-M perpetual archives over [2024-08-01, 2026-08-01). The project began with a stricter premise: trade, mark, index, and OI had to occupy one exact five-minute grid continuously for at least 365 days. The checksum-closed archive audit falsified that premise. The longest un-repaired intersection contained 87,717 rows, or 304.5729166666667 days. We preserved that negative finding and revised the admission rule rather than silently repairing the data. The revised core consists of trade, mark, index, and realized funding aligned by availability time; OI is optional because its publication time is unverified.

This study asks:

1. Can the revised core streams support at least 365 complete point-in-time days and a frozen temporal split?
2. Does a deterministic auditor correctly classify frozen legal and known-violation templates?
3. Does auditing reduce false passes in null-signal controls?
4. Does an adaptive audited proposal policy outperform random search and tree GP under identical valid-candidate budgets?
5. Do frozen candidates retain predictive and economic value in a one-time historical holdout?

Our contributions are deliberately narrow. First, we report both the failed exact-grid premise and the revised availability-masked admission result. Second, we integrate three-timestamp rules, executable-plan auditing, null controls, append-only candidate accounting, and one-time holdout access. Third, we report the negative search and economic findings rather than selecting a post-holdout rescue specification. We do **not** claim the first safe, auditable, or agentic factor-mining framework; accepted work already covers formula search, reinforcement learning, LLM factor agents, and structure-aware exploration (Ren et al. 2024; Yu et al. 2023; H. Shi et al. 2025; Li et al. 2024; Chen et al. 2026; Y. Shi, Duan, and Li 2026).

2. Related Work

RiskMiner uses risk-seeking Monte Carlo tree search for formulaic alpha discovery (Ren et al. 2024). AlphaGen studies reinforcement learning for synergistic formula collections (Yu et al. 2023), while AlphaForge jointly mines and dynamically combines formulaic factors (H. Shi et al. 2025). FAMA evaluates a neural-symbolic LLM factor-mining agent (Li et al. 2024). AlphaSAGE uses structure-aware GFlowNets and emphasizes exploration, convergence, runtime, and cross-universe evidence (Chen et al. 2026). Navigating the Alpha Jungle combines LLM guidance with MCTS and rich search feedback (Y. Shi, Duan, and Li 2026). These studies motivate strong baselines, repeated runs, out-of-sample evaluation, ablation, and economic assessment. Their principal evidence is equity-centric and does not establish the specific public-archive timing semantics tested here.

Recent public work also narrows any broad novelty claim. Constrained cryptocurrency factor agents, safe DSL execution, executable program evolution, high-frequency digital-asset factor feedback, and execution-governed perpetual tuning have all been publicly described (Huang et al. 2026; R. Shi et al. 2026; Lin et al. 2026; Zhang et al. 2026; Deng 2027). Our distinguishing question is consequently not whether an agent can emit formulas, but whether a perpetual-specific point-in-time audit changes false passes and whether an adaptive proposer survives an exactly matched valid-evaluation comparison.

3. Method

3.1 Data source and market-level fallback

Historical OKX probes returned old trade, mark, and index candles but empty old funding and OI responses through the tested public route. The frozen fallback rule permitted a whole-study switch, not cross-venue stream mixing. We therefore used public Binance BTCUSDT USD-M archives (Binance 2026). The raw manifest contains 826 official-checksum-verified ZIP files: 24 monthly files for each of trade, mark, index, and funding, plus 730 daily metrics/OI files.

3.2 Event, publication, and availability time

For a bar, event time identifies the interval, publication cannot precede the explicit close, and availability is `close_time + 1 ms`. For realized funding, event time is settlement time. The archive lacks a distinct publication-time field, so the primary analysis uses event time plus five minutes as a research assumption, with 0/5/10/15-minute admission sensitivity. A decision at time `t` may read only records with `availability_time <= t`.

Missing observations are removed through an explicit mask. Interpolation, backward filling, cross-gap forward filling, timestamp snapping, deletion of conflicting duplicates to manufacture continuity, and venue mixing are prohibited. OI

does not determine admission and is disabled as a factor input because its publication time is unverified.

3.3 Protocol revision and split

The retired rule measured the exact five-minute intersection of trade, mark, index, and OI. The revised rule measures complete decision days for trade, mark, index, and native-frequency funding after availability masking. These are different sample definitions, not two estimates of one quantity.

The revised data produced 727 complete UTC days. A timestamp-only 60/20/20 split yielded 436 training days, 145 validation days, and 146 test days, with 60-minute purge and embargo. Because the protocol was hash-locked after the historical interval had ended, the last partition is called a **one-time historical holdout**, not a prospective test.

3.4 Deterministic auditor

Search methods propose expressions, but a deterministic evaluator exclusively parses the expression, constructs lineage, applies availability masks, audits execution, computes labels and metrics, performs selection, and writes the ledger. The auditor rejects five known classes:

1. future source shift;
2. same-bar execution;
3. future realized funding;
4. incorrect OI availability alignment; and
5. backward fill.

The conformance benchmark contains 40 illegal and 40 legal executable templates, with eight violations per class. Acceptance required at least 38/40 illegal templates detected, no more than 2/40 legal templates rejected, and 8/8 exact reason matches in every class. Wilson intervals accompany, but do not replace, this finite-template rule.

3.5 Null-signal benchmark

The benchmark uses five moving-block label permutations of public 15-minute BTC data and five synthetic heavy-tailed, volatility-clustered null paths. Each path evaluates 100 candidates, including 80 legal and 20 known-violation templates. We compare no audit, a basic shift audit, and the full five-class audit under 2%, 5%, and 10% missingness and three policies: exact-grid deletion, prohibited naive backward fill, and availability masking. Selection uses BH-FDR $q=0.10$; PBO and Deflated Sharpe diagnostics are also recorded. The primary comparison is 5% missingness with availability masking.

3.6 Matched-budget search

All methods share an expression DSL with maximum AST depth 4, at most 5 operator nodes, and lookbacks of 3/6/12/24/48/96/288 five-minute bars. A numerically valid candidate must pass the audit, be unique, have at least 0.95 finite coverage, and have finite train and validation IC. Later qualification requires BH-FDR $q < 0.10$, a dependence-aware interval excluding zero, and performance above the random-search 90th percentile, followed by $|\text{correlation}| < 0.8$ deduplication.

We compare:

- **audited agent:** a deterministic adaptive proposal policy using training-only failure and validation feedback; it is not an external LLM;
- **random:** uniform proposals in the same DSL;
- **tree GP:** subtree mutation and crossover under the same DSL and validity gate.

The seeds are 11, 23, 37, 53, and 71; horizons are 15 and 60 minutes. Each method–seed–horizon run evaluates 100 valid candidates, giving 1,000 valid candidates per method. All successful, invalid, duplicate, and rejected proposals are retained.

3.7 Frozen holdout and economic evaluation

At most five qualified factors per run are direction-corrected on validation data, z-scored, and equally weighted. Position is the sign of the composite, delayed by one completed decision bar and sampled non-overlapping at the target horizon. The primary cost model uses a conservative Binance research assumption of 6 bp taker fee per side plus 2 bp slippage per side. Fee sensitivity is 4/5/6/8/10 bp and slippage sensitivity is 1/2/5/10 bp per side. Realized funding is charged only when a position crosses a settlement event.

The holdout evaluator was invoked once after selection and economic rules were frozen. Candidate-level IC intervals use 500 moving-block resamples. Post-holdout method summaries use a descriptive 10,000-resample seed-cluster bootstrap that retains both horizons for a sampled seed.

4. Experiments

We organize the experiments by the five research questions.

- **Data audit:** census coverage, duplicates, ordering defects, exact-grid gaps, funding cadence, and lag sensitivity.
- **Template audit:** confusion matrix and reason-code accuracy on 80 frozen programs.
- **False-discovery controls:** 10 primary null paths and 270 audit-by-gap scenarios.

- **Search comparison:** 3 methods $\times$ 5 seeds $\times$ 2 horizons $\times$ 100 valid candidates.
- **Historical holdout:** frozen candidates, primary costs, 20 fee-slippage combinations per evaluated run, one-bar delay, extreme-observation deletion, and volatility-state checks.

The principal outcomes are complete days; violation recall and legal false rejection; false-pass rate and PBO; valid and qualified candidates; test IC and RankIC; net Sharpe, cumulative return, drawdown, turnover, and robustness counts. Direction accuracy, trade count, and evaluations to first discovery were not emitted upstream and are not reconstructed.

5. Results

5.1 Public-archive availability

Table 1: Native-stream archive coverage and point-in-time admission roles.

Stream	Admission role	Rows/events	Dates	Availability rule
Trade	Core	210,240	730	close + 1 ms
Mark	Core	209,952	729	close + 1 ms
Index	Core	209,952	729	close + 1 ms
Funding	Core	2,190	730	event + 5 min assumption
OI	Optional; disabled	210,235	730	unverified

Trade had no exact-grid gap. Mark and index each lacked 288 bars on 2026-06-29. Funding had no separate publication-time field. OI had 9 missing and 3 off-grid records, one conflicting duplicate group, and 501 archive-order reversals; these defects were retained rather than repaired.

The old trade/mark/index/OI exact-grid requirement failed at 304.5729166666667 continuous days. Under the revised core definition, the five-minute primary mask retained 209,951 eligible decisions, masked 288, and yielded 727 complete UTC days. Funding lag 0/5/10/15 minutes left the complete-day count at 727; longest eligible runs were 697.000/697.000/696.997/696.993 days. This supports data admission only, not alpha.

5.2 Known-template audit

All 40 rejection reasons matched the injected class. The point estimates pass the frozen template rule, but the intervals do not establish population recall above 95% or false rejection below 5%. The generator directly exercises known auditor rules, so unknown leakage is outside the evidence.

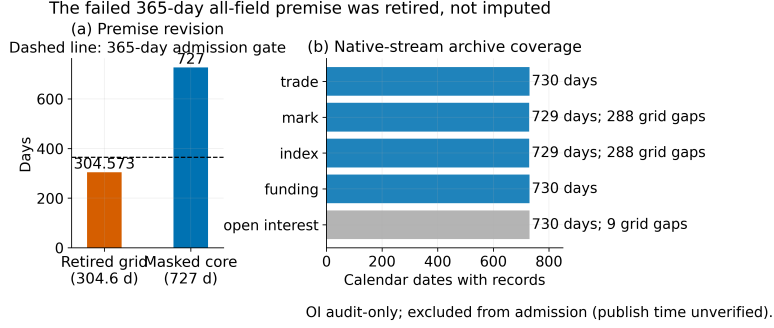


Figure 1: Retired exact-grid and revised masked-core availability (different sample definitions); the dashed line marks the 365-day admission gate

Table 2: Known-template auditor conformance with Wilson uncertainty.

Scope	Detected or rejected	Rate	Wilson 95% interval
Illegal templates	40/40	1.000	[0.9124, 1.0000]
Legal templates falsely rejected	0/40	0.000	[0.0000, 0.0876]
Each violation class	8/8	1.000	[0.6756, 1.0000]

5.3 Null controls and missing-policy ablation

At 5% missingness with availability masking, mean false passes were 0.2910 without auditing, 0.2479 with the basic audit, and 0.0625 with the full audit. Full versus no audit improved all 10 paths; the paired difference was -0.2285 with a path-bootstrap 95% interval [-0.2595, -0.2012] and exact two-sided sign-test $p=0.001953$. The relative reduction was 78.5%.

Conditional holdout-failure differences were uncertain because full auditing often selected zero or few candidates. Failure burden per evaluated candidate fell by 0.1062, interval [-0.1237, -0.0865]. PBO changed by only -0.001429, interval [-0.004286, 0], with nine ties. DSR was zero where computable and missing when no candidate was selected.

Table 3: Mean false-pass rates at 5% missingness by audit and missing-data policy.

Audit	Exact-grid drop	Backward fill	Availability mask
None	0.2910	0.2890	0.2910
Basic	0.2479	0.2521	0.2479
Full	0.0625	0.0525	0.0625

Exact-grid deletion and availability masking were identical in the executed simulation, while backward fill was not uniformly worse. Therefore the supported

result is the joint effect of the full known-rule audit pipeline, not an independent effect of masking.

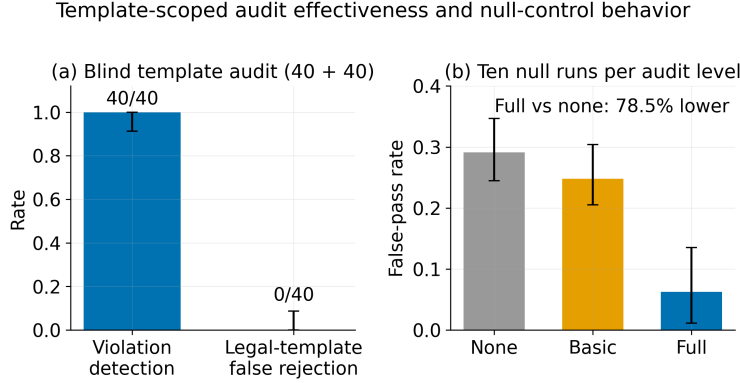


Figure 2: Template audit and null controls

5.4 Search accounting and efficiency adjudication

Table 4: Matched-budget search accounting and qualification yield.

Method	Runs	Attempts	Valid	Qualified	Runs	Yield/100 valid
					selected	(95% CI)
Audited agent	10	1,648	1,000	39	8	3.9 [1.9, 5.0]
Random	10	2,658	1,000	39	8	3.9 [1.9, 5.0]
Tree GP	10	1,804	1,000	29	8	2.9 [1.1, 4.7]

The audited agent tied random search on qualified candidates and therefore did not beat both baselines. Raw attempts were lower for the agent, but evaluations to first qualified factor and resource-normalized runtime were not frozen outcomes. We reject search-superiority and general efficiency claims.

5.5 Historical-holdout prediction and economics

Validation-to-test IC sign was retained in all 23 evaluated runs, and individual moving-block IC intervals excluded zero. Nevertheless, no evaluated run had positive net Sharpe at primary costs. An additional one-bar delay produced 0/23 positive Sharpe, and no run remained economically positive after the frozen extreme-observation deletions.

The fee-slippage grid contained 160 agent, 160 random, and 140 GP cells. Positive-Sharpe counts were 0, 0, and 0. The best observed Sharpe values were -6.3148, -5.5009, and -5.4938, respectively. Thus all 460 executed cost cells were negative. Positive IC did not imply an economically usable strategy.

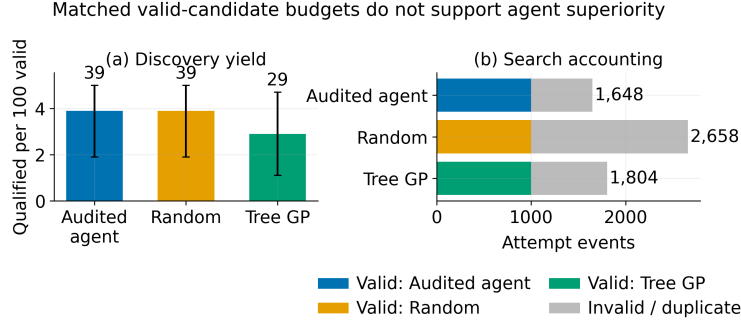


Figure 3: Matched-budget search accounting: blue, orange, and green denote matched-valid audited-agent, random, and tree-GP candidates; gray denotes invalid or duplicate attempts

Table 5: Historical-holdout predictive and economic results at primary costs.

(a) *Predictive results*

Method	Runs (eval./none)	Mean test IC (95% CI)	Mean RankIC
Audited agent	8/2	0.2352 [0.1551, 0.3552]	0.2136
Random	8/2	0.1907 [0.1164, 0.2681]	0.1826
Tree GP	7/3	0.1550 [0.1262, 0.1765]	0.1375

(b) *Economic results*

Method	Positive Sharpe	Median Sharpe	Mean DD	Turnover
Audited agent	0/8	-28.04	-0.9291	6,235.9
Random	0/8	-21.16	-0.9460	6,652.8
Tree GP	0/7	-30.11	-0.9008	5,158.1

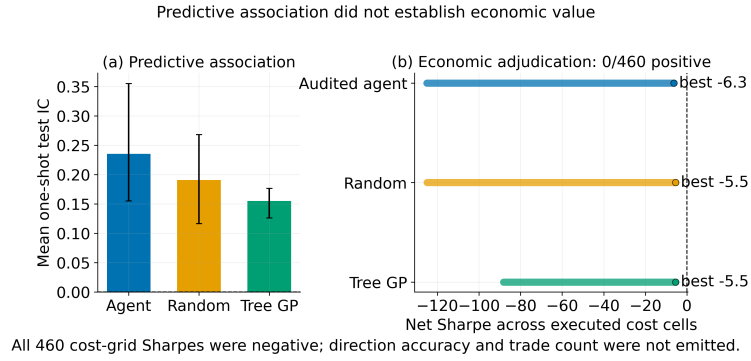


Figure 4: Predictive and economic adjudication

6. Discussion

The strongest positive result is procedural: known-rule auditing removed candidates that should not enter statistical selection and sharply reduced false passes

in the executed null controls. The result is relevant because archive completeness alone concealed both an invalid original data premise and concrete OI and bar gaps. Yet the benchmark is not an unseen-adversary test, and its finite-template intervals are wide.

The experiment also separates prediction from economics. Positive holdout IC coexisted with very high turnover and uniformly negative cost-adjusted results. Changing the combiner, holding period, or cost model after opening the holdout could have produced a more attractive retrospective specification, but would invalidate the one-time-access design. We retained the negative result.

The adaptive proposer’s tie with random search is likewise informative. Its lower raw-attempt count does not establish the preregistered discovery-efficiency claim, especially because first-discovery and resource measurements were not emitted. The result applies only to this deterministic adaptive policy, DSL, data, and budget; it is not evidence that all LLM or agentic search is ineffective.

7. Limitations

1. **Single asset and venue.** Evidence is limited to Binance BTCUSDT USD-M and cannot establish cross-market validity.
2. **Retrospective lock.** The holdout interval ended before the protocol lock; one-time access is not prospective validation.
3. **Funding-time assumption.** The archive has no separate publication timestamp. The 5-minute assumption was tested for admission, not predictive outcomes.
4. **OI excluded.** OI publication time is unverified and its archive contains gaps, off-grid records, a conflict, and ordering reversals.
5. **Template scope.** The audit benchmark covers known injected rules, not unknown or adversarial leakage implementations.
6. **Gap-mechanism identification.** The null simulation did not distinguish exact-grid deletion from availability masking.
7. **Incomplete search curves.** Budgets 30 and 300, first-discovery counts, and a resource-normalized comparison were not executed.
8. **Missing baseline.** LightGBM was unavailable; the broader planned non-linear and manual baselines were not completed.
9. **Execution approximation.** Aggregated bars and assumed slippage do not support capacity or live-tradability claims. Direction accuracy and trade count were not emitted.
10. **No external validation.** There is no second asset, exchange, or genuinely future interval.

8. Conclusion

A file-complete public archive was not automatically a point-in-time-complete research dataset. The original exact-grid requirement failed at 304.5729166666667 days; a disclosed, scientifically different core-stream definition yielded 727 com-

plete days without filling. The deterministic auditor passed a finite known-template benchmark and reduced false passes across the executed null controls, but the missing-policy ablation did not identify an independent masking effect. Under matched valid-candidate budgets, the audited adaptive proposer tied random search and did not establish superiority. Positive historical-holdout IC failed every primary and sensitivity economic test.

The evidence supports an audit-focused negative-result paper, not a profitability or general agent-superiority claim. Detailed protocols, additional tables, artifact hashes, and exact reproduction commands are provided in the supplement and reproducibility record.

Binance. 2026. “Binance Public Data.” Public archive. <https://data.binance.vision/>.

Chen, Binqi, Hongjun Ding, Ning Shen, Jinsheng Huang, Taian Guo, Luchen Liu, and Ming Zhang. 2026. “AlphaSAGE: Structure-Aware Alpha Mining via GFlowNets for Robust Exploration.” In *The Fourteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=zRKF4ln2VE>.

Deng, Kaihong. 2027. “AutoQuant: An Auditable Expert-System Framework for Execution-Constrained Auto-Tuning in Cryptocurrency Perpetual Futures.” *Expert Systems with Applications* 333: 133924. <https://doi.org/10.1016/j.eswa.2026.133924>.

Huang, Yikuan, Zheqi Fan, Kaiqi Hu, and Yifan Ye. 2026. “From Hypotheses to Factors: Constrained LLM Agents in Cryptocurrency Markets.” <https://arxiv.org/abs/2604.26747>.

Li, Zhiwei, Ran Song, Caihong Sun, Wei Xu, Zhengtao Yu, and Ji-Rong Wen. 2024. “Can Large Language Models Mine Interpretable Financial Factors More Effectively? A Neural-Symbolic Factor Mining Agent Model.” In *Findings of the Association for Computational Linguistics: ACL 2024*, 3891–3902. Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-acl.233>.

Lin, Qinhong, Ruitao Feng, Yinglun Feng, Zhenxin Huang, Yukun Chen, Zhongliang Yang, Linna Zhou, Binjie Fei, Jiaqi Liu, and Yu Li. 2026. “FactorEngine: A Program-Level Knowledge-Infused Factor Mining Framework for Quantitative Investment.” <https://arxiv.org/abs/2603.16365>.

Ren, Tao, Ruihan Zhou, Jinyang Jiang, Jiafeng Liang, Qinghao Wang, and Yijie Peng. 2024. “RiskMiner: Discovering Formulaic Alphas via Risk Seeking Monte Carlo Tree Search.” In *Proceedings of the 5th ACM International Conference on AI in Finance*, 752–60. Association for Computing Machinery. <https://doi.org/10.1145/3677052.3698613>.

Shi, Hao, Weili Song, Xinting Zhang, Jiahe Shi, Cuicui Luo, Xiang Ao, Hamid Arian, and Luis Angel Seco. 2025. “AlphaForge: A Framework to Mine and Dynamically Combine Formulaic Alpha Factors.” *Proceedings of the AAAI Conference on Artificial Intelligence* 39 (12): 12524–32. <https://doi.org/10.1609/aaai.v39i12.33365>.

- Shi, Runze, Shengyu Yan, Yuecheng Cai, and Chengxi Lv. 2026. “Hubble: An LLM-Driven Agentic Framework for Safe, Diverse, and Reproducible Alpha Factor Discovery.” <https://arxiv.org/abs/2604.09601>.
- Shi, Yu, Yitong Duan, and Jian Li. 2026. “Navigating the Alpha Jungle: An LLM-Powered MCTS Framework for Formulaic Alpha Factor Mining.” *Proceedings of the AAAI Conference on Artificial Intelligence* 40 (2): 997–1005. <https://doi.org/10.1609/aaai.v40i2.37069>.
- Yu, Shuo, Hongyan Xue, Xiang Ao, Feiyang Pan, Jia He, Dandan Tu, and Qing He. 2023. “Generating Synergistic Formulaic Alpha Collections via Reinforcement Learning.” In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 5476–86. Association for Computing Machinery. <https://doi.org/10.1145/3580305.3599831>.
- Zhang, Lingzhe, Tong Jia, Yunpeng Zhai, Zixuan Xie, Chiming Duan, Minghua He, Philip S. Yu, and Ying Li. 2026. “From Feedback Loops to Policy Updates: Reinforcement Fine-Tuning for LLM-Based Alpha Factor Discovery.” <https://arxiv.org/abs/2605.15412>.